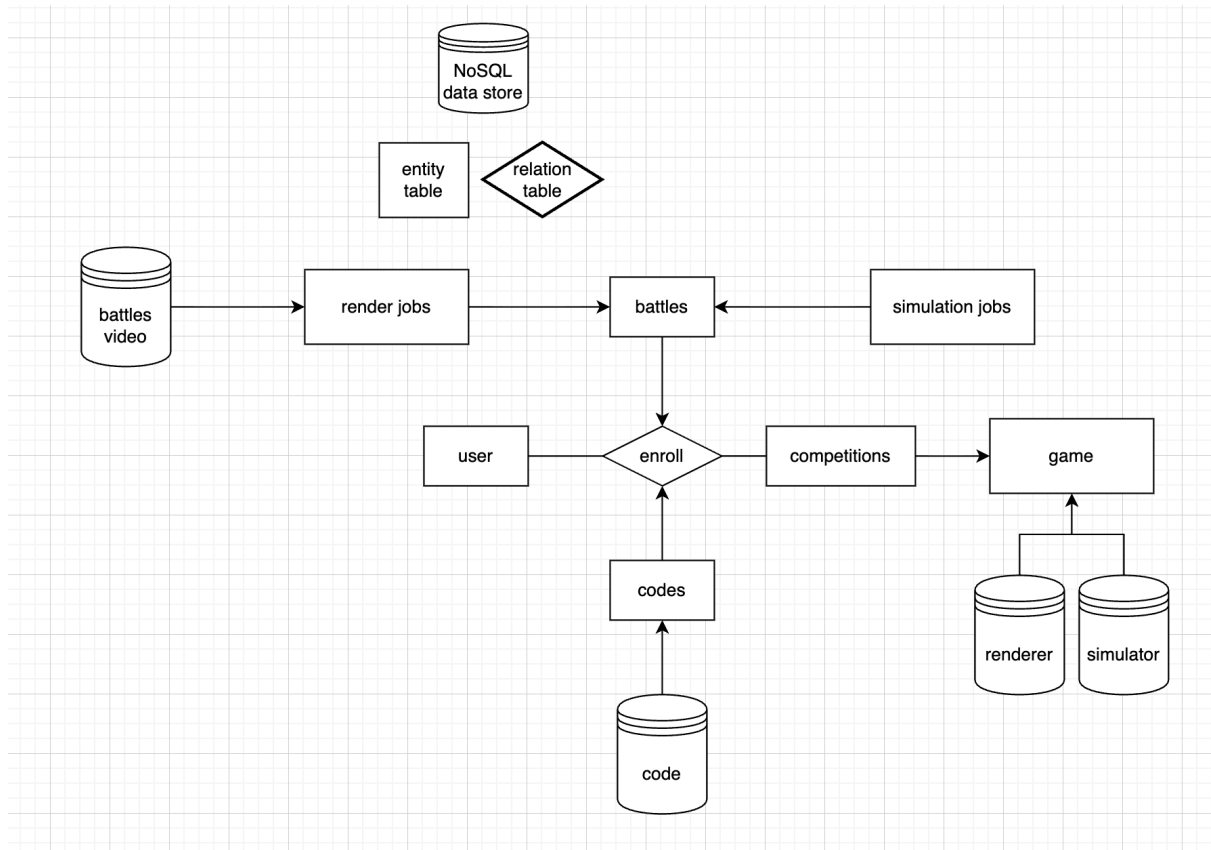


YYC Coding Platform - System Architecture & Design

1. Data model



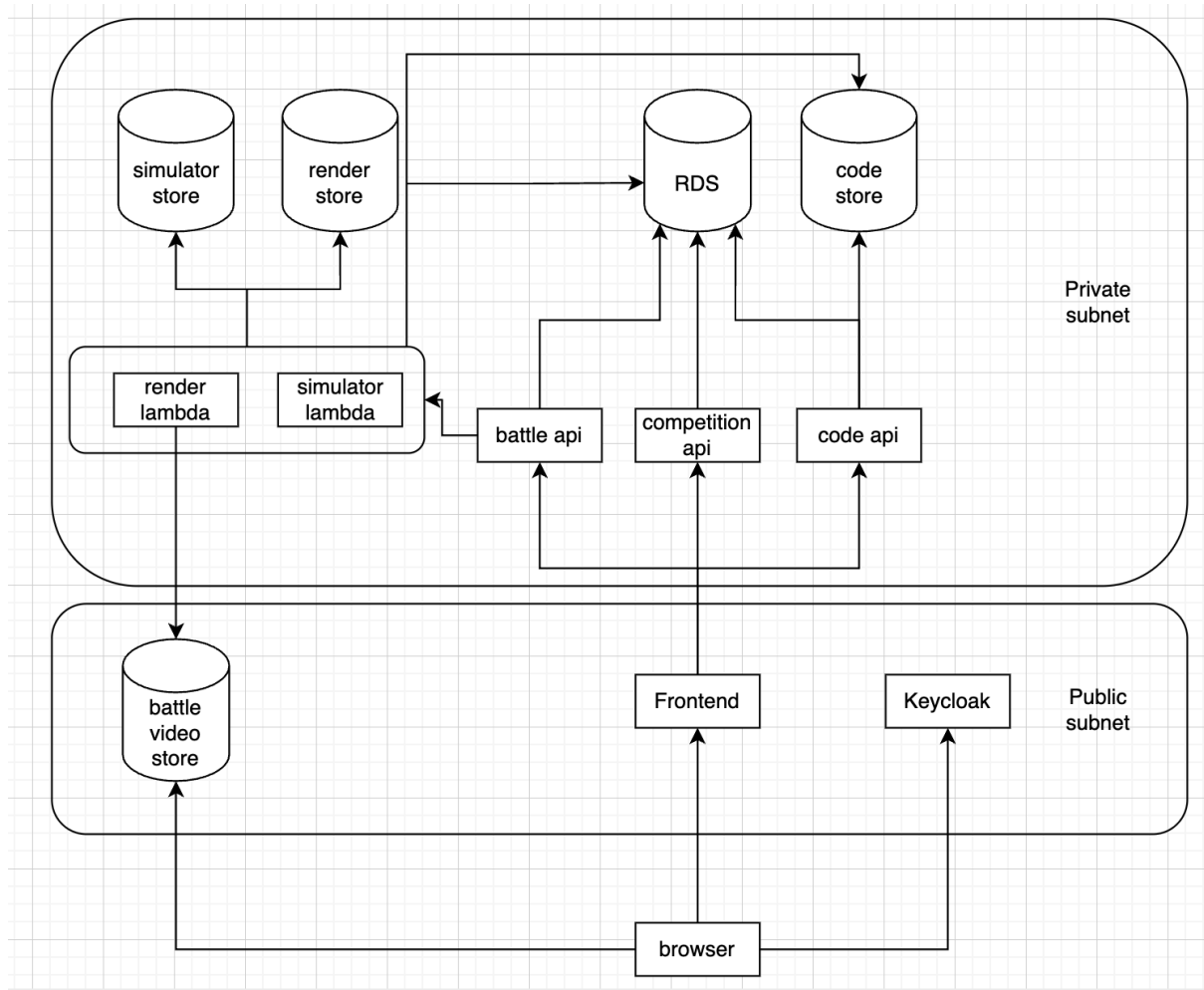
2. Database design

game	- id - name - display_name - simulator_reference - renderer_reference - created_at_utc - updated_at_utc	- uuid - varchar(20) - varchar(20) - varchar(256) - varchar(256) - timestamp - timestamp		
competition	- id - game_id - name - display_name - start_time_utc - end_time_utc - enabled	- uuid - uuid - varchar(20) - varchar(20) - timestamp - timestamp - bool		
user	- id - keycloak_id - username - full_name - nickname - created_at_utc - updated_at_utc	- uuid - uuid - varchar(20) - varchar(50) - varchar(20) - timestamp - timestamp		
enroll	- id - competition_id - user_id - selected_code_id	- uuid - uuid - uuid - uuid		
codes	- id - enroll_id - name - code_reference	- uuid - uuid - varchar(20) - varchar(50)		
battles	- id - a_enroll_id - b_enroll_id - a_code_id - b_code_id - created_at_utc - updated_at_utc	- uuid - uuid - uuid - uuid - uuid - timestamp - timestamp		
simulation_jobs	- id - battles_id - status - winner_user_id - winner_score_gain - loser_score_loss - created_at_utc - updated_at_utc	- uuid - uuid - pending / complete / fail - uuid - float - float - timestamp - timestamp		
render_jobs	- id - battles_id - status - battle_video_reference - created_at_utc - updated_at_utc	- uuid - uuid - pending / complete / fail - varchar(50) - timestamp - timestamp		

TODO:

1. 究竟simulator 同render step 使唔使2合1？對response time影響有幾大？
2. 如果唔2合1, 咁就一定得save低intermediate output, 會多一個s3 folder.

3. Architecture design



4. Alibaba Cloud VS AWS

https://www.alibabacloud.com/en/product/product-mapping?_p_lc=1

AWS	Alibaba Cloud	Reference
Cloudformation	ROS	https://www.alibabacloud.com/help/en/resource-orchestration-service/latest/getting-started?spm=a2c63.l28256.0.0.2ee1489ez99TEg
RDS PostgreSQL	ApsaraDB RDS for PostgreSQL	
S3	OSS	
Lambda	Function Compute (FC)	
EventBridge	Eventbridge	
EKS	ECS	